Ryan Compton

Contact Information

ryan@ryancompton.net https://ryancompton.net

Employment

Staff Software Engineer, Machine Learning, Google. September 2017–present. Work on ranking for Google Discover, a recommendation feed used by hundreds of millions of people. Designed a new experiment system and framework for the combiner function on top of our multi-head ranking network. Did diversity and quality work in the packing stage, including making sure quality breaking news reaches the feed. Built an agentic triage system for user feedback.

Head of Applied Machine Learning, Clarifai. June 2015–September 2017. Built, fine-tuned, and deployed convnets to solve computer vision problems for some of our most important verticals and largest customers. With company expansion took on management responsibilities and oversaw team growth from scratch to 10 direct reports.

Research Staff Member, Howard Hughes Research Laboratories. June 2012–June 2015. Developed algorithms for mining massive modern datasets and predictive analytics. Conceived of and implemented a system which accurately inferred home locations for over one hundred million Twitter users and combined it with multilingual natural language processing tools to generate forecasts for thousands of international events.

Education

MS, PhD, Mathematics, UCLA, 2006–2012

Dissertation: Sparsity promoting optimization in quantum mechanical signal processing.
Advisor: Chris Anderson

BA, Mathematics/Physics, New College of Florida, 2002–2006

Thesis: Optimizing cover times with constraints.
Advisor: Patrick McDonald

Selected first-author publications

Ryan Compton, David Jurgens, David Allen, "Geotagging One Hundred Million Twitter Accounts with Total Variation Minimization" *IEEE BigData 2014* arXiv:1404.7152
Press coverage: Forbes, Business Insider, New York Observer, Daily Caller, MIT Technology Review, Komando, Serene Risc Digest, Schneier on Security

Ryan Compton, Craig Lee, Tsai-Ching Lu, Lalindra De Silva, Michael Macy, "Detecting future social unrest in unprocessed Twitter data: Emerging phenomena and big data", *IEEE-ISI*, 2013 **best paper nomination**

Ryan Compton, Stanley Osher and Louis Bouchard, "Hybrid regularization for MRI reconstruction with static field inhomogeneity correction", *IEEE International Symposium on Biomedical Imaging*, May 2012 (Journal version: Inverse Probl. Imag. 7, 1215–1233 (2013))

Selected talks

What Convnets Look at When They Look at Nudity
Summary: Overview of how deconvnets can be used to visualize representations learned by a convolutional neural network trained to filter nude photos.
Details: Blog, video and deck.
Venues: NYC ML, GDG NYC, Boston ML, ODSC 2017, DevFest DC 2017 (keynote)

Geotagging One Hundred Million Twitter Accounts with Total Variation Minimization
Summary: Technical presentation of a social network analysis method which can geolocate the overwhelming majority of active Twitter users, independent of their location sharing preferences, using only publicly-visible data.
Details: Paper and deck.
Venues: IEEE BigData, UCLA Level Set, UCI ISG, UCR CS, LMU

Computer skills

Programming: Yes. Most recently C++ and Python. LaTeX res.cls file.
Operating Systems: Linux user at home since 2002. I was also the admin for a 24-node Hadoop/Spark cluster at HRL.

Work authorization

United States and European Union